tests of other features of general relativity on the Earth, in the solar system, and in binary systems that contain compact, exotic stars called pulsars. General relativity has come through each test with flying colors.

Over the past thirty years I have participated in the theoretical-physics quest which produced our present understanding of black holes and in the quest to test black-hole predictions by astronomical observation. My own contributions have been modest, but with my physicist and astronomer colleagues I have reveled in the excitement of the quest and have marveled at the insight it has produced. This book is my attempt to convey some sense of that excitement and marvel to people who are not experts in either astronomy or physics.

---

1. Chapters 3, 6, 7.

2. Chapter 8

3. Chapter 8.

4. Chapters 3 and 6.

5. Chapter 6.

6. Chapter 2.

7. Readers who want to compute properties of black holes for themselves will find the relevant formulas in the notes at the end of the book.

8. Chapters 3 –5.

9. Chapter 2.

10. For further discussion of the concept that the laws of physics *force* black holes, and the solar system, and the Universe, to behave in certain ways, see the last few paragraphs of Chapter 1.

11. Chapter 2.

12. Chapter 7.

13. Chapter 7.

14. Ibid.

15. Chapter 3. The quantity 18.5 kilometers, which will appear many times in this book, is 41t (that is, 12.5663706 . . .) times Newton's gravitation constant times the mass of the Sun divided by the square of the speed of light. For this and other useful formulas describing black holes, see the notes to this chapter.

16. Chapter 13.

17. Chapters 3 and 13.

18. Chapter 13

19. See Box 2.3.

20. Chapters 2 and 3